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Article in *Clinical and Experimental Ophthalmology* · December 2017

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Review

Retinal pigment epithelium in the pathogenesis of age-related macular degeneration and photobiomodulation as a potential therapy?

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ABSTRACT

The retinal pigment epithelium (RPE) comprises a monolayer of cells located between the neuroretina and the choriocapillaries. The RPE serves several important functions in the eye: formation of the blood-retinal barrier, protection of the retina from oxidative stress, nutrient delivery and waste disposal, ionic homeostasis, phagocytosis of photoreceptor outer segments, synthesis and release of growth factors, reisomerization of all-trans-retinal during the visual cycle, and establishment of ocular immune privilege. Age-related macular degeneration (AMD) is the leading cause of blindness in developed countries. Dysfunction of the RPE has been associated with the pathogenesis of AMD in relation to increased oxidative stress, mitochondrial destabilization and complement dysregulation. Photobiomodulation or near infrared light therapy which refers to non-invasive irradiation of tissue with light in the far-red to near-infrared light spectrum (630–1000 nm), is an intervention that specifically targets key mechanisms of RPE dysfunction that are implicated in AMD pathogenesis. The current evidence for the efficacy of photobiomodulation in AMD is poor but its safety profile and proposed mechanisms of action motivate further research as a novel therapy for AMD.

Key words: age-related macular degeneration (AMD), low light therapy, near infrared light, photobiomodulation, retinal pigment epithelium (RPE).

INTRODUCTION

The retinal pigment epithelium (RPE) comprises a monolayer of cells located between the retinal photoreceptors and the fenestrated choriocapillaris, which is basally bordered by the elastin and collagen-rich Bruch's membrane (BM). It serves several important functions in maintaining photoreceptor health. Its tight junctions protect the neuroretina from blood-borne toxins and its light-absorbing pigments mitigate photo-oxidative stress. The RPE delivers nutrients and disposes waste products including phagocytosis of photoreceptor outer segments (POS). It is fundamental to the visual cycle by recycling of retinoids. Furthermore, the RPE contributes to the production and secretion of growth factors and provides ocular immune privilege.

Age-related macular degeneration (AMD) accounts for 8.7% of worldwide blindness and is the leading cause of blindness in developed countries.¹ Although the pathogenesis of AMD remains incompletely understood, there is evidence that dysfunction of the RPE resulting from oxidative stress, impairment of nutrient delivery as well as energy metabolism and complement dysregulation, contributes to the pathogenesis of AMD.

Photobiomodulation, also known as near infrared (NIR) light therapy, is known to increase cellular metabolism, energy supply and metabolic repair processes.^{2,3} Hence, photobiomodulation of dysfunctional RPE could conceivably restore function and attenuate the pathogenesis of AMD.

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Received 11 September 2017; accepted 28 November 2017.

Conflict of interest: None declared

Funding sources: None declared.

The aim of this review is to summarize the structure and functions of the RPE and its role in the pathogenesis of age-related macular degeneration, and to discuss the current evidence and potential of photobiomodulation in AMD.

STRUCTURE

The RPE comprises a polygonal monolayer of cells, the size and shape of which vary as a function of retinal eccentricity, from a diameter of 14 μm at the fovea to considerably larger (60 μm) in the peripheral retina.^{4–9} The height of RPE cells is a 10–15 μm at the fovea and approximately 7.5 μm at the periphery.¹⁰ The cell density of the RPE is greater in the fovea compared to the equator.¹¹ With age, peripheral RPE density declines, but this is preserved at the fovea due to inward migration of peripheral RPE.¹² Approximately 30–40 photoreceptors overlie each RPE cell.¹¹ The RPE projects microvilli apically that enshroud the POS. Long thin apical microvilli measuring 5–7 μm in length are projected from the RPE and enshroud the POS, facilitating transepithelial and retina-to-RPE transport. Similarly, the basal surface of the RPE contains complex infoldings to support molecular movement. This apical-basal polarity is also reflected in the organization of RPE organelles and membrane proteins. Melanosomes are situated apically whilst the nucleus, mitochondria, endoplasmic reticulum, lysosomes, Golgi apparatus, lipofuscin and melanolipofuscin granules are located towards the basal side.^{7,13} The polar distribution of different combinations of ion channels and pumps on either the apical or basal membrane enables the diffusion of certain ions in specific directions in order to transport nutrients to photoreceptors and remove retinal waste products as well as the maintenance of subretinal ionic homeostasis as elucidated below.

BLOOD-RETINAL BARRIER

Tight junctions between adjacent RPE cells are attached to the actin cytoskeleton and form part of the outer blood-retinal barrier.¹⁴ This prevents paracellular transport of large molecules, toxins, blood-borne products and water.¹⁵ As a result of the tight junctional barrier produced by the RPE, resistance to paracellular movement is 10 times higher than resistance to transcellular movement.^{16,17} However, this resistance is plastic, as permeability of the tight junctions can change in the presence of diffusible factors produced from the neural retina.¹⁸ One such factor is vascular endothelial growth factor (VEGF) which has been implicated in the breakdown of the outer blood-retinal barrier in diabetic macular oedema.^{19,20} Hypoxia and hyperglycaemia may

upregulate hypoxia inducible factor 1 α and therefore VEGF expression, resulting in altered tight junction proteins.^{20–22} During the development of the RPE, tight junctions tend to be leaky, but these become progressively less permeable due to secretion of these diffusible factors from the neural retina.^{18,23}

PROTECTION FROM OXIDATIVE DAMAGE

The RPE is responsible for the absorption of reflected and scattered light. This helps to both optimize image quality and protect the retina from oxidative damage caused by reactive free radicals, which result from the combination of constant RPE exposure to light, the high RPE metabolic activity, the elevated local oxygen tension and the photo-oxidation of lipofuscin.^{6,7} The pigment melanin, which is stored and synthesized in melanosomes, absorbs and filters harmful light, particularly in the blue wavelength region of the spectrum which is known to cause photo-oxidation of lipofuscin, a lipid-protein pigment that accumulates with age and from phagocytosis of the POS.^{24,25} Melanin has been identified as having additional antioxidative properties, including counteracting singlet oxygen and scavenging reactive oxygen species (ROS).^{26,27} Melanin density increases towards the centre of the retina, reaching its peak at the fovea, contributing to the clinically darker appearance of the fovea and surrounding macula. Melanin within the RPE decreases uniformly after the age of 40.⁷ Another mechanism of defence against oxidative damage is the intrinsic antioxidant defence system of the RPE, which comprises both enzymes, such as catalase and superoxide dismutase, and non-enzymatic antioxidants such as α -tocopherol, β -carotene and ascorbate.^{28–32} Glutathione has been implicated as another antioxidant in the RPE with functions including peroxidase-reduction of lipid hydroperoxides, maintenance of ascorbate in its reduced form as well as detoxification of reactive products of lipid peroxidation such as 4-hydroxynonenol.^{33,34}

TRANSPORT OF NUTRIENTS, WASTES AND WATER

The close anatomical apposition of the RPE to the photoreceptors is necessary for facilitating the transcellular movement of nutrients, wastes, fluid and ions.

The delivery of nutrients, including glucose, ascorbate and fatty acids, from the choroidal vasculature to the photoreceptors occurs through transporters within the RPE membranes. Glucose is passively transported through numerous GLUT1 (glucose transporter 1) and GLUT3 (glucose transporter 3) transporters on both

the apical and basal membranes.^{35,36} More recently, additional glucose transporters GLUT2 (glucose transporter 2) and GLUT5 (glucose transporter 5) have been identified on the apical and basolateral membranes in cultured human RPE cells.³⁷

Ascorbate is a carbohydrate vitamin that is vital in hydroxylation reactions and biosynthetic pathways, especially in the production of collagen and glycosaminoglycans which are components of the interphotoreceptor matrix (IPM) and BM.^{38–40} It is also an effective antioxidant and scavenger of superoxide radicals. The transcellular movement of ascorbic acid across the RPE occurs through active sodium-dependent transport that is reduced by presence of ouabain, 2,4-dinitrophenol and dehydroascorbic acid.⁴⁰

Photoreceptor membranes, which comprise a significant portion of the omega 3 fatty acid docosahexanoic acid (DHA), are continuously renewed and added to the POS as photoreceptor tips are shed and phagocytized by the RPE.⁴¹ Thus, photoreceptors require a constant supply of DHA. Initially, DHA is produced in the liver from the dietary precursor α -linolenic acid before being carried by lipoproteins in the circulation to the RPE. Transport through RPE membranes occurs through a concentration-dependent manner.^{41–43} DHA acid has also been identified as a precursor to neuroprotectin 1, which protects RPE from oxidative damage and may also play a role in RPE retinoid transport by modulating binding between interphotoreceptor binding protein (IRPB) and 11-cis-retinal.⁴⁴

Lactic acid, a metabolic end product produced in excess by the retina and thought to be specifically released from POS and to a lesser extent, photoreceptor inner segments has been proposed to act as a nutrient for RPE.^{45,46} Analysis of bovine eyes revealed the distribution of lactate on the retinal surface was seven times greater than at the RPE whilst glucose was 50 times greater at the RPE than the retinal surface. The distribution of these fuels suggests RPE consumes more lactate and releases more glucose from the retina.⁴⁶ However no comparison has been made in vivo and the mechanism of lactate metabolism in RPE remains unclear. From the subretinal space, lactic acid enters the RPE apical membrane by active transport through lactate-H⁺ co-transporter monocarboxylate transporter 1 (MCT1) and the Na⁺-dependent co-transporter. Irrespective of whether lactate acts as a fuel source for RPE cells or not, excess levels of this compound are removed through the RPE basolateral membrane via monocarboxylate transporter 3 (MCT3) and the Na⁺-lactate exchanger.^{47,48}

Constant transport of water away from the subretinal space to the choriocapillaris is required to maintain close apposition of the RPE and photoreceptors.

Large volumes of subretinal fluid are generated from metabolic turnover in photoreceptors as well as intraocular fluid flux from the vitreous body towards the retina. Water travels across the RPE to the choriocapillaris by active transport through the aquaporin 1 channel, identified in studies of cultured rat and human RPE.^{49,50} Water transport is also facilitated by the transport of ions such as Cl[−] and K⁺ which will be elaborated below.

TRANSEPITHELIAL TRANSPORT AND MAINTENANCE OF SUBRETINAL ION LEVELS

The RPE plays a major role in preserving excitability of photoreceptors by careful regulation of subretinal ionic concentration through transepithelial transport of ions (Fig. 1). The apical Na⁺/K⁺ ATPase pump, which results in outflow of Na⁺ and influx of K⁺, provides the energy gradient that is vital to transepithelial transport.^{51–53} The Na⁺/K⁺ ATPase, together with the polarized distribution of other channels and transporters, produce a high subretinal sodium concentration and high potassium concentration in the RPE cytosol. The elevated subretinal sodium is critical to the dark current which is produced by the entry of sodium ions through cGMP (cyclic guanosine monophosphate) gated channels, resulting in depolarization of photoreceptors. Apical entry of sodium occurs through the Na⁺–K⁺–2Cl[−] co-transporter and exit occurs basolaterally through mechanisms that are unclear.⁵⁴ However, most intracellular sodium ions are recycled through the apical Na⁺/K⁺ ATPase to maintain the high subretinal sodium concentration.

The RPE also effectively compensates for ionic changes such as fluctuation in potassium levels during phototransduction. Potassium can enter the RPE apically through Na⁺/K⁺ ATPase or Na⁺–K⁺–2Cl[−] co-transporter and leave basolaterally or apically through K⁺ channels.⁵⁴ Whilst in the dark, the potassium concentration is 5 mM with net transepithelial efflux of potassium from the subretinal space to the choroid. Exposure to light causes the hyperpolarization of photoreceptors leading to the closure of cGMP gated Na⁺ channels in the outer segment membranes and reduced K⁺ efflux from the inner segment. Subsequently, the subretinal potassium concentration decreases to 2 mM which triggers hyperpolarization of the RPE apical membrane and activation of inward rectifying K⁺ channels on the apical membrane, causing K⁺ flow into the subretinal space. In the early phase of RPE hyperpolarization, the apical Na⁺–K⁺–2Cl[−] co-transporter reverses direction, further increasing K⁺ movement to the subretinal space, resulting in the restoration of subretinal potassium concentration.⁵⁵

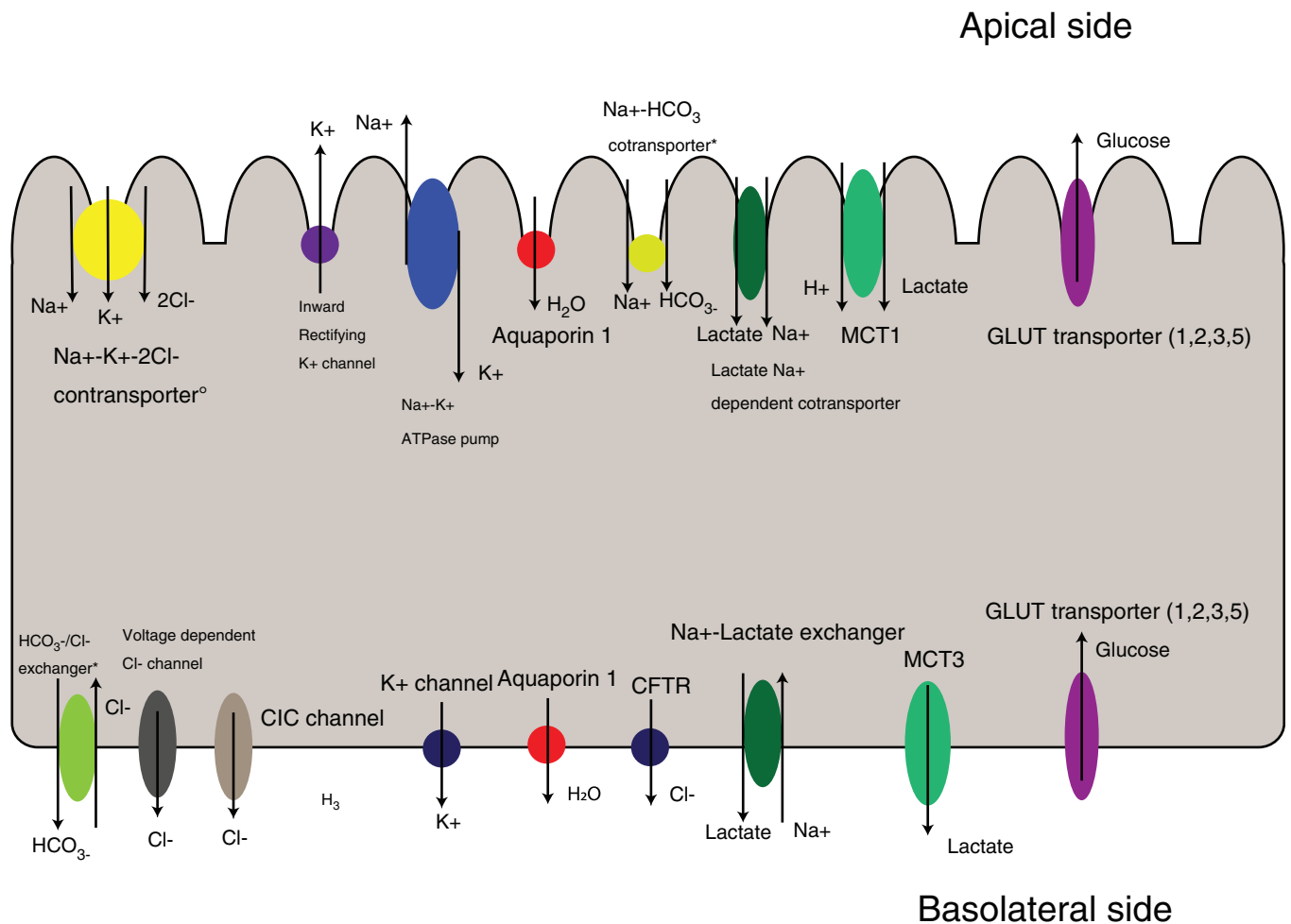


Figure 1. Glucose enters and exits through GLUT transporters 1,2,3 and 5. Lactate enters apically through MCT1 and lactate Na^+ co-transporter before exiting basolaterally via MCT3 and Na^+ lactate exchanger. The apical Na^+/K^+ ATPase pump, which results in out-flow of Na^+ in exchange for K^+ , provides the energy gradient that for transepithelial transport. Sodium enters apically through $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ co-transporter and exits basolaterally through mechanisms that are unclear. Potassium can enter apically through Na^+/K^+ ATPase or $\text{Na}^+-2\text{K}^+-2\text{Cl}^-$ co-transporter and leave basolaterally or apically through K^+ channels. Chloride enters the RPE apically via the $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ co-transporter and exits basolaterally via voltage-dependent chloride and CIC-2 channels. The basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchanger, which is involved in pH regulation, facilitates chloride entry into the RPE cell. CFTR channels on the basolateral aspect facilitates the exit of chloride. Transport of water occurs through aquaporin channels distributed apically and basolaterally.

Chloride, a vital facilitator for fluid movement, enters through the RPE apical membrane via the $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ co-transporter before exiting basolaterally via voltage-dependent chloride and CIC-2 channels. The basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchanger, an important channel in pH regulation, transports chloride back into the RPE. An additional efflux channel for chloride in RPE cells is the CFTR channel on the basolateral aspect.⁵⁴ CFTR channels along with the CIC family channels have been found to be highly susceptible to oxidative stress in human foetal and adult RPE cultures.^{56,57} Chloride channel dysfunction and altered fluid transport secondary to oxidative stress may play a role in AMD.⁵⁸ Chloride transport, similar to potassium transport in the RPE is regulated by calcium, cAMP (cyclic adenosine monophosphate), epinephrine and ATP (adenosine triphosphate).^{57,59}

Transepithelial transport of HCO_3^- is intrinsically associated with pH regulation of the subretinal space and the RPE cell itself. At high pH, HCO_3^- travels apically into the RPE via the $\text{Na}^+/\text{HCO}_3^-$ transporter and leaves basolaterally in exchange with Cl^- through the $\text{HCO}_3^-/\text{Cl}^-$ exchanger. Conversely, at low pH HCO_3^- enters basolaterally through the $\text{HCO}_3^-/\text{Cl}^-$ exchanger before travelling to the subretinal space through the $\text{Na}^+/\text{HCO}_3^-$ transporter.⁶⁰

PHAGOCYTOSIS OF POS

Photoreceptor membranes undergo a renewal process whereby the outer segment tips are shed and subsequently phagocytosed by RPE before new outer segments are constructed at the cilium. This is

necessary for several reasons: to maintain photoreceptor excitability, to recycle nutrients, including DHA and retinal and to prevent oxidative damage from photo-oxidation of damaged POS lipid and protein components.

The process of POS phagocytosis is triggered by interactions with ligand receptors including CD36 (Cluster of Differentiation 36),^{61,62} the mannose-6-phosphate receptor,⁶³ alpha V beta5 integrin,⁶⁴ MerTK (C-MER proto-oncogene Tyrosine Kinase),^{65,66} CD81 (cluster of differentiation 81)⁶⁷ and L-type calcium channels.⁶⁸ Initially, specific binding of the POS to the RPE apical membrane leads to induction of a secondary messenger cascade. This results in the ingestion of the POS before undergoing lysosomal digestion. The components from lysosomal digestion are either recycled back to photoreceptors as nutrients or transported into the choroid vasculature as waste.

PRODUCTION AND SECRETION OF GROWTH FACTORS

In order to preserve the neural retina and choroid architecture, the RPE produces and secretes several protein mediators, cytokines and growth factors, including platelet-derived growth factor (PDGF), pigment epithelium-derived growth factor (PEDF), vascular endothelial growth factor, fibroblast growth factors (FGF), transforming growth factor β (TGF β), insulin-like growth-I (IGF-I), ciliary neurotrophic factor (CNTF), tissue inhibitor of metalloprotease (TIMP), lens epithelium-derived growth factor (LEDGF) and members of the interleukin family.^{69–75} Photoreceptors are protected by the neuroprotectant growth factors PEDF, FGF, CNTF, LEDGF and IGF-I whilst PDGF modulates cell growth and healing. PEDF, which is secreted from the RPE apical membrane, has anti-angiogenic properties and helps maintain the fenestrations of the choriocapillaris endothelium. TGF β moderates inflammation and regulates extracellular matrix synthesis and turnover, a function shared by TIMP. VEGF, which is secreted from basal RPE, basically has an opposite role to PEDF and promotes angiogenesis and regulates the permeability of the choriocapillaris. VEGF overexpression has been implicated as a key player in the pathogenesis of choroidal neovascularisation (CNV) in AMD.^{76,77} The expression of VEGF is regulated by numerous factors, including mechanical stress, hypoxia advanced glycation end products (AGE), cytokines (interleukin 1 and TNF), vasopressor hormones (angiotensin II), vasopressin, growth factors (TGF β , FGF and PDGF) and L-type Ca²⁺ channels.⁷⁸

REISOMERIZATION OF ALL-TRANS-RETINAL AND TRANSPORT OF RETINOIDS IN THE VISUAL CYCLE

The visual cycle is the fundamental process by which a photon is converted into an electrical signal in the retina. The triggering event of the visual cycle is the absorption of light by rhodopsin in the POS which results in one of its components, the chromophore 11-cis-retinal, to be converted to all-trans-retinal. This results in a conformational change in opsin, a G-protein coupled receptor component of rhodopsin, which activates the regulatory protein transducin to initiate signal transduction cascades leading to closure of cyclic GMP-gated cation channels, hyperpolarization of the photoreceptor cell membrane and, ultimately, decreased glutamate release. The RPE plays an integral role in the transport of retinoids as well as reisomerization of all-trans-retinal to 11-cis-retinal in the visual cycle. Following light exposure, all-trans-retinal is converted into all-trans-retinol by retinal dehydrogenase in the POS before being transported to the RPE by interstitial retinol-binding protein 3 (RBP3).^{79,80} All-trans-retinol can also originate from the blood circulation, being carried by serum retinol-binding protein, before entering the RPE cell basolaterally through the STRA6 (stimulated by retinoic acid 6) protein.^{81,82} Whilst in the RPE, all-trans-retinol is bound to cellular retinol-binding protein and esterified to all-trans-retinyl-ester in the presence of the catalyst lecithin retinol acyl transferase. All-trans-retinyl-ester undergoes isomerization through RPE65 to produce 11-cis-retinol before being converted by 11-cis-retinol dehydrogenase to produce 11-cis-retinal. 11-cis-retinal is then transported by RBP3 to the POS where it combines with opsin to reform rhodopsin. The cycle is repeated following light exposure whereby the resulting all-trans-retinal is reconverted to its all-trans-retinol and cycled back to the RPE for reisomerization and oxidation.^{8,81}

IMMUNE PRIVILEGE

The RPE has a central role in maintaining immune privilege in the eye. Firstly, the outer blood-retinal barrier formed by tight junctions between RPE cells creates a microenvironment which allows infiltration into the retina by immune system components to be carefully regulated. Secondly, the RPE itself is capable of secreting immunosuppressive factors, including TGF β , interleukin 11 and interferon β , to downregulate T-cell activity.⁸³ Thirdly, the presence of Fas ligand on the membrane allows RPE cells to induce apoptosis in Fas expressing effector

leukocyte cells.^{84,85} Furthermore, RPE cells which express major histocompatibility complex (MHC) class I and II are able to act as antigen-presenting cells in the eye.^{86,87} RPE cells can also synthesize or express numerous complement proteins and regulators. PCR analysis of RPE cells revealed that the RPE was capable of local production of complement 3(C3), complement factor B (CFB), complement factor H (CFH), complement factor D (CFD) and complement factor I (CFI). Complementary regulatory proteins membrane cofactor protein (MCP), decay accelerating factor (DAF) and CD59 are also expressed on the RPE membrane.⁸⁸

AGE-RELATED MACULAR DEGENERATION

AMD is an ocular disease associated with aging and characterized symptomatically by impairment of central vision. Classically, AMD is divided into two subtypes: 'early' and 'late' AMD. In the early stages, AMD is marked by increasing and abnormal deposition of extracellular debris between Bruch's membrane and the RPE, known as drusen. Yellowish subretinal lesions known as subretinal drusenoid deposits or pseudodrusen located initially in the outer superior macula are markers for progression of AMD.^{89,90} The progression of early AMD is characterized by large drusen formation and/or clinically visible RPE abnormalities. Late AMD is characterized by the development of geographic atrophy and increasing impairment of central vision. There is currently no treatment for the atrophic form of AMD. Neovascular AMD is the other clinical variant of late AMD and causes more rapid and dramatic loss of central vision and is characterized by chorioidal neovascularisation, the development of new blood vessels from the choroid and rarely the retina that often leak into the retina, causing haemorrhage, retinal detachment and eventually disciform scars. Anti-VEGF intravitreal injections are currently the main form of treatment for CNV. Although the pathogenesis of AMD remains unclear, there is evidence that degeneration or dysfunction of RPE cells often occurs in the early AMD pathogenesis and can be a predictor of photoreceptor loss although in cases such as outer retinal atrophy associated with pseudodrusen, photoreceptor loss can precede RPE disease.⁹¹⁻⁹⁵

OXIDATIVE STRESS

Oxidative stress has been suggested as a potential integral mechanism of RPE injury in the pathogenesis of AMD. *In vitro* experiments where RPE cells are exposed to oxidative stress, that is, H_2O_2 have shown increased apoptosis, decreased proliferation and features of senescence.⁹⁶⁻⁹⁸ Apoptotic RPE cells

have also been identified in CNV membranes from AMD donor eyes.⁹⁹

The underlying mechanism of oxidative stress-induced apoptosis in AMD is thought to be an interference in the protective epidermal growth factor receptor/protein kinase B pathway as well as an activation of apoptosis-promoting factors such as caspase 3 and 9, as triggered by the release of cytochrome c from mitochondria.^{97,100} AMD donor eyes have histological evidence of established widespread oxidative damage in patients with advanced GA and also in eyes with CNV to a lesser extent.¹⁰¹ Genetic evidence of the importance of RPE oxidative stress in the pathogenesis of AMD is supported by transgenic mice having a deletion of the superoxide dismutase 1 (SOD 1) gene. These mice develop clinical features of AMD, including drusen deposition, RPE atrophy and CNV.¹⁰²

Carboxyethyl pyrole (CEP) residues, which are oxidized remnants of DHA, have been used as biomarkers in patients with AMD who have increased plasma CEP autoantibodies and accumulation of CEP within the RPE and outer retina.¹⁰³ Mice that have been immunized with mouse serum albumin adducted with CEP developed AMD-like lesions similar to GA.¹⁰⁴ Subretinal injections of CEP modified human serum albumin have also been found to exacerbate angiogenesis in laser-induced CNV mouse models.¹⁰⁵

A major source of oxidative stress in RPE cells is the formation of ROS during phagocytosis of POS. Extracellular H_2O_2 is presumably generated from the action of NAPH (nicotinamide adenine dinucleotide phosphate) oxidase in a phagosome or from peroxidation of ingested POS lipids. Downregulation of the antioxidant enzyme catalase also occurs during phagocytosis, leading to increased susceptibility of RPE cells to ROS.²⁹

Lipofuscin, autofluorescent collections of undegradable protein and lipids associated with aging, accumulate in lysosomes of postmitotic cells including RPE cells. Lipofuscin within RPE cells has been postulated to originate from insufficient RPE phagocytosis of POS and it is thought that this may also contribute to the decline in RPE phagocytic function with aging.¹⁰⁶ N-retinyl-N-retinylidene ethanolamine (A2E), a component of lipofuscin, is theoretically susceptible to photo-oxidation from exposure to blue light, generating toxic A2E-epitopes that cause oxidative damage and subsequent caspase 3-mediated apoptosis of RPE cells.^{107,108} However, the role of A2E in the pathogenesis of AMD has recently been questioned. Recent mass spectrometry studies of human RPE have shown that greater amounts of A2E were located in the periphery rather than the central region, incongruent with the spatial distribution of lipid fluorescence.^{109,110} In addition, clinical

evaluation of lipofuscin in early AMD patients showed no significant increase compared to age-matched controls.¹¹¹

Epidemiological studies associating cigarette smoking and sunlight to increased risk of developing AMD support the role of oxidative stress as a key factor in the pathogenesis of AMD.¹¹² Wang *et al.*¹¹³ demonstrated exposure to benzo(a)pyrene, the toxic element in cigarette smoke, resulted in apoptosis of cultured RPE cells as well as ultrastructural changes akin to oxidative damage in mice.

The role of oxidative stress in AMD pathogenesis is also supported by the AREDS study which demonstrated high-dose antioxidants (vitamin C, vitamin E and β -carotene) as well as zinc, which increases the activity of enzymes such as SOD and catalase, significantly reduced the progression of intermediate AMD to advanced AMD.¹¹⁴ The follow-up AREDS2 study in 2013 concluded that the antioxidants lutein and zeaxanthin were effective and safe alternatives to β -carotene which had, as a side-effect, been found to increase lung cancer in smokers.¹¹⁵

MITOCHONDRIAL DYSFUNCTION

Impaired mitochondrial function in the RPE appears to be a critical factor in AMD. Mitochondria are an endogenous source of oxidative stress. Mitochondrial injury or dysfunction results in impaired respiration which leads to the increased accumulation of ROS.¹¹⁶ In a continuous cycle, oxidative stress in mitochondria can exacerbate the generation of ROS, eventually overwhelming RPE cells resulting in apoptosis. Mitochondrial DNA (MtDNA) is a more vulnerable target to oxidative damage than nuclear DNA (nDNA) given its proximity to the ROS-producing inner mitochondrial membrane, relatively poor repair capabilities and its lack of protective histones or other DNA-associated proteins. Furthermore, the coding regions of MtDNA are more predisposed to oxidative damage since there exist a large number of intronless regions with high transcription rates.¹⁰⁰ *In vitro* experiments, where RPE cells were exposed to oxidative stress demonstrated preferential damage to MtDNA over nDNA along with a decline in mtDNA repair and compromised mitochondrial redox function.^{98,117} In human AMD donor eyes, there is evidence of increased macula-specific mtDNA damage, mitochondrial heteroplasmic mutations and diminished mtDNA repair capacity in the RPE.¹¹⁸ The association between AMD and RPE mitochondrial dysfunction was further illustrated in a proteomic analysis of AMD donor eyes demonstrating altered mitochondrial translation factors, decreased ATP synthase subunits and reduced mitochondrial heat shock protein 70 expression

(mtHsp70).¹¹⁹ Important functions of mtHSP70 include regulation of p53-mediated apoptosis, iron-sulphur cluster biogenesis, mitochondrial calcium regulation and ATP-dependent transport of nuclear-encoded proteins into the mitochondrial matrix.¹¹⁹

A2E in lipofuscin has been shown to induce apoptosis of RPE cells in a light-dependent manner through a mitochondria-related mechanism. Upon exposure to blue light, A2E binds non-covalently to cytochrome oxidase C (COX), an essential mitochondrial enzyme in oxidative phosphorylation. Upon attachment, A2E exhibits its toxic effect by inhibiting COX causing impaired mitochondrial respiration, generation of ROS and formation of proapoptotic complexes involving cytochrome C.^{120–122}

There is also genetic evidence for the association between mitochondrial dysfunction and AMD. In human AMD patients, mtDNA lesions were increased significantly in all regions of the mitochondrial genome compared to age-matched controls where age-related mtDNA lesions only occurred in the common deletion regions.¹²³ In addition, polymorphisms of ARMS2 gene, which encodes for an outer mitochondrial protein in RPE, is associated with significantly increased predisposition to AMD.^{124,125} Macular dystrophy patients resulting from the A3243G mtDNA mutation display clinical phenotypes similar to AMD including varying levels of RPE atrophy and subretinal deposits.^{126,127}

COMPLEMENT DYSREGULATION

Inappropriate activation of the complement system has been implicated in AMD pathogenesis. Biochemical analysis of drusen constituents reveals a significant number of complement activators (A2E, amyloid β , immunoglobulins, CRP, advanced glycation end products and cholesterol), complement components and inhibitors.⁸⁸

The alternate complement pathway (AP) has been the main complement pathway implicated in AMD pathogenesis. Although there have been reports of classical pathway involvement, with variants of C2 genes and SERPING1 gene encoding c1 inhibitor being associated with AMD, this has not been confirmed in larger case-controlled studies.^{128–130} Genetic variations of the genes encoding AP proteins CFH (inhibitor of AP),^{131–133} CFB¹²⁸, CFI^{134,135} as well as C3¹³⁶ component are associated with higher susceptibility to AMD, indicating the significance of AP dysregulation in AMD pathogenesis. Haplotypes of CFH with deletion of CFH-related proteins protects against AMD.¹³⁷ Genetic mutations associated with altered extracellular matrix such as the EFEMP1 and TIMP3 mutations have been associated with inherited macular diseases, displaying local inflammation as well as

complement factor deposition. Patients display AMD characteristics such as sub-RPE drusen-like deposits as well as eventual atrophy and CNV.^{138,139} Rodent models with genetic knockout of CFH display structural and functional retina degenerations similar to AMD. Increasing expression of human CFH in this model resulted in the inhibition of AP complement pathway, reduction of sub-RPE deposits as well as prevention of retinal and kidney damage caused by CFH deletion in a dose-dependent manner.^{140,141} The process of AP-mediated RPE cell death which occurs in GA involves lysis of cells marked with C3b by membrane attack complex (MAC) that is modulated by extracellular calcium.¹⁴²

The AP also has a critical role in the modulation of CNV in laser-induced models.¹⁴³ Complement components C3a, C5a, CFB and MAC induce CNV by upregulating RPE secretion of angiogenic factors including VEGF, TGF- β 2 and β -fibroblast growth factor.^{144,145} Consequently, specific inhibition of AP reduced angiogenesis in CNV mouse models.¹⁴⁶

The complement system may have a role in promoting the chronic local inflammatory process in AMD. C3a and C5a are known to have chemotactic properties and can increase RPE expression of the inflammatory cytokines interleukin-1 β , interleukin-6, interleukin-8, MCP-1 and granulocyte-macrophage colony-stimulating factor (GM-CSF).¹⁴⁷

There is evidence that oxidative stress can render RPE more susceptible to complement-mediated damage.^{142,148} RPE cells exposed to oxidative stress exhibited reduced expression of the membrane bound complement inhibitors DAF, CD55 and CD59 and downregulation of CFH.^{149–151} Additionally, mice immunized with CEP display increased deposition of C3d below the RPE.¹⁰⁴ In cultured human RPE cells, complement and oxidative stress synergistically increased VEGF secretion up to 100-fold.¹⁴⁹

PHOTOBIMODULATION

Photobiomodulation refers to the non-invasive irradiation of tissue with light in the far-red to near-infrared light spectrum (630–1000 nm) with delivery methods varying from laser sources to light emitting diode (LED) devices.¹⁵² Photobiomodulation originated in the 1960s shortly after the advent of the laser. Mester *et al.*¹⁵³ were the first to demonstrate the positive effects of photobiomodulation. Using NIR lasers they were able to induce increased hair growth and wound healing in mice and treat non-healing skin ulcers in human patients. Since then the field of photobiomodulation has broadened to include treatment of a range of conditions including wound healing, diabetic ulcers, neurological

pain, peripheral nerve injury, stroke and myocardial infarction.¹⁵⁴ More recently, focus has turned to the potential beneficial effects of photobiomodulation as a treatment for blinding retinal diseases. Treatment effects from photobiomodulation has been demonstrated in various retinal disease animal models including light-induced retinal degeneration,^{155,156} AMD,¹⁵⁷ retinitis pigmentosa,¹⁵⁸ diabetic retinopathy¹⁵⁹ and retinopathy of prematurity^{160,161} as evidenced by improved ERG, decreased inflammatory markers and diminished cell loss.

Photobiomodulation has been reported to have positive results in patients with AMD,^{162,163} diabetic retinopathy,¹⁶⁴ amblyopia¹⁶⁵ and retinitis pigmentosa,¹⁶⁶ reflected in improvement of visual acuity and decreased visual field loss. However, these clinical studies range from interventional case series to a single case report. In addition, the mechanism of photobiomodulation particularly on amblyopia is unclear. Hence, the current evidence for the efficacy of photobiomodulation in retinal disease is poor. Nevertheless the mechanism of photobiomodulation and its relative safety should motivate more robust clinical studies.

MECHANISMS OF PHOTOBIMODULATION

In the far-red and near-infrared light spectrum, COX is the primary photoacceptor. By targeting COX, photobiomodulation modulates electron transfer in the reduction of oxygen during mitochondrial respiration, hence increasing the mitochondrial membrane potential and ATP synthesis (Fig. 2). Ultimately, this triggers and enhances cellular repair processes and metabolism in photoreceptors, choroid and retinal pigment epithelium.¹⁶² Evidence of COX as the primary target comes from Eels who reported photobiomodulation improved rat retinal function following methanol intoxication which is known to inhibit cytochrome oxidase activity.¹⁶⁷ NIR light treatment of primary cultured neurones reversed the effect of tetrodotoxin by upregulating COX activity.¹⁶⁸ Desmettre *et al.*¹⁶⁹ found in choroidal layers, that transpupillary application of laser therapy-induced increased expression of heat shock proteins, which are known to stimulate cellular metabolism and prevent premature cell death. Another potential mechanism of photobiomodulation is the unbinding of nitric oxide (NO) from COX as demonstrated by Karu *et al.*¹⁷⁰ Since NO inhibits mitochondrial respiration, its dissociation from COX would restore mitochondrial oxygen consumption, therefore increasing energy production and boosting cellular metabolic processes.¹⁵² *In vitro* experiments have reported that photobiomodulation therapy increases phagocytosis and lysosomal activity, cellular processes important in the reduction of

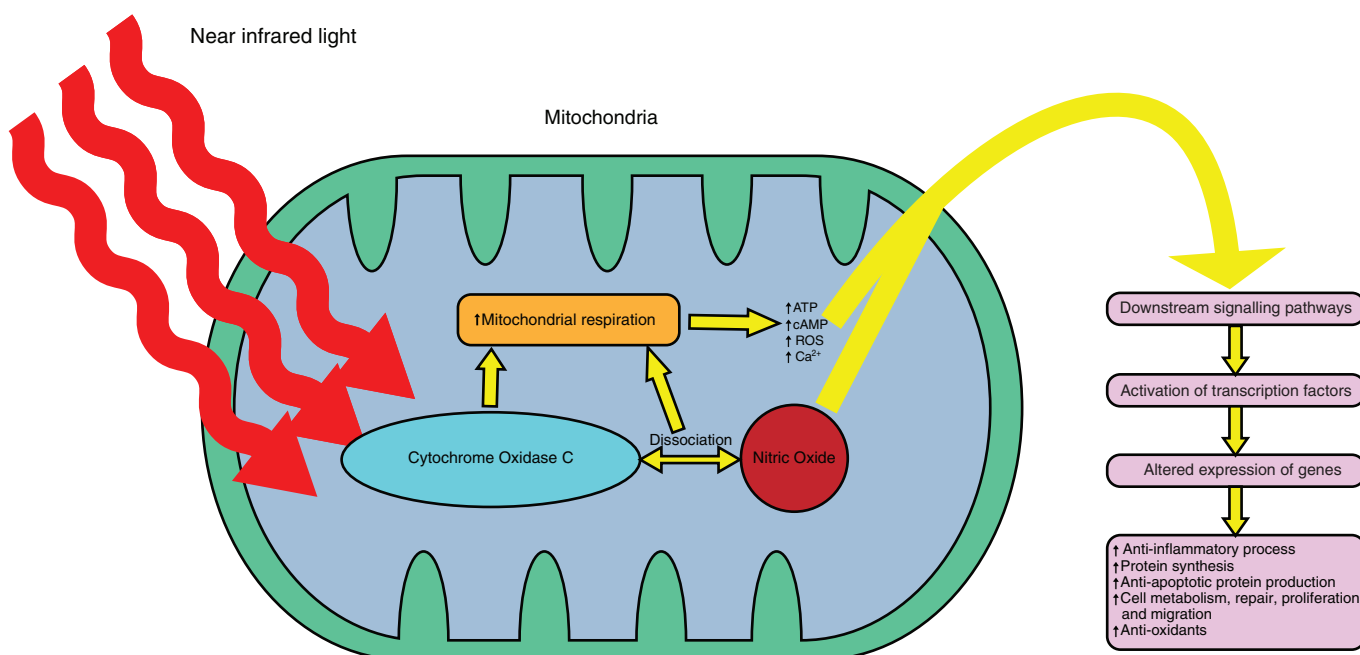


Figure 2. Schematic diagram of the mechanisms of photobiomodulation near infrared light within wavelengths (630–1000 nm) targets the mitochondrial enzyme cytochrome oxidase C resulting in (i) direct stimulation in mitochondrial respiration and (ii) dissociation of nitric oxide which indirectly increases mitochondrial respiration. These processes result in elevation of ATP, cAMP, reactive oxygen species and intracellular calcium which impact downstream signalling pathways that triggers increase in anti-inflammatory processes, protein synthesis, anti-apoptotic protein production, cellular repair/metabolism/proliferation/migration and antioxidants.

inflammation and enhancing repair of the retina.¹⁵⁴ These processes occur from photobiomodulation triggering downstream signalling cascades via ATP, cAMP, NO, ROS and Ca^{2+} resulting in eventual gene expression that promotes protein synthesis, anti-inflammatory processes, antioxidants, anti-apoptotic proteins as well as cell migration and proliferation.¹⁷¹ Photobiomodulation usually stimulates mitochondrial ROS production at low levels however it may decrease ROS production during oxidative stress.¹⁷² In addition, photobiomodulation has observed to have a biphasic dose response *in vitro* in particular relation to ATP levels and mitochondrial potential.¹⁷³

APPLICATIONS IN AMD

The pathogenesis of AMD as discussed above encompasses a complex interplay of oxidative stress, mitochondrial dysfunction and complement dysregulation, likely involving the RPE. Photobiomodulation may prove to be an effective treatment for AMD as there is evidence that it can significantly modulate all of these pathological processes (Table 1.).

Exposure of NIR light with a dose of 2.88 J/cm^2 to human RPE cells *in vitro* stimulated a 56% increase in ATP and twofold increase in intracellular NO production at 5 h post-exposure, presumably via unbinding of NO from COX leading to an

enhancement in mitochondrial oxidative phosphorylation.¹⁷⁴ Photobiomodulation resulted in a six-fold increase in levels of growth promoting nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B), an 11-fold increase in levels of apoptosis suppressor protein Bcl-2 and a 70% decrease in levels of the apoptotic effector protein Bax. In RPE cultures exposed to the oxidative stressor H_2O_2 and challenged with POS, Fumar *et al.*¹⁷⁵ reported improvement of phagocytosis, increased expression of MerTK and reduced ROS production following treatment with 250 s of 670 nm light at 3.89 mW/cm^2 for four consecutive days. Interestingly, mitochondrial membrane potential was not affected by NIR light in this study. Photobiomodulation has also been demonstrated to protect RPE cells from the lethal effects of thermal laser.¹⁷⁶

There is also considerable evidence of a beneficial effect of photobiomodulation in animal AMD models. In an aged mice model of AMD, Kokkinopoulos¹⁷⁷ reported that photobiomodulation could improve mitochondrial function of RPE cells and reduce inflammation. Following five treatments of NIR light exposure at 40 mW/cm^2 lasting 90 s, there was significantly increased mitochondrial membrane potential as well as reduced component C3d, proinflammatory cytokine TNF- α and reduced macrophage numbers. This was confirmed by Begum *et al.*¹⁵⁷ who reported similar results in CFH knock-out mice. Mice exposed to NIR light at 20 mW/cm^2

Table 1. Applications of photobiomodulation in AMD

Experiment	Delivery	Dose	Frequency	Results
hTERT-RPE culture (Lavey <i>et al.</i>) ¹⁷⁴	671 and 637 nm light via LED, unspecified duration	2.88 J/cm ²	Unspecified	• x2 ↑NO • 56% ↑ATP 5 h post • x6 ↑NF-κB • x11 ↑BCl-2 • 70% ↓Bax
Human RPE cultures exposed to H ₂ O ₂ and POS (Fumar <i>et al.</i>) ¹⁷⁵	670 nm light via LED for 250 s	3.89 mW/cm ²	Twice daily for 4 days	• ↑phagocytosis • ↑MerTK expression • ↓ROS production
C57BL/six mice (Kokkinopoulos <i>et al.</i>) ¹⁷⁷	670 nm light via LED for 90 s	40 mW/cm ²	Five times over 35 h	• ↑mitochondrial membrane potential • ↓C3d • ↓TNF-α • ↓macrophage numbers • ↑COX • ↓C3 complement deposition in outer retina • ↓vimentin and GFAP • ↑ATP
CFH knockout mice (Begum <i>et al.</i>) ¹⁷⁸	670 nm light via LED environmental lighting for 6 min	20 mW/cm ²	Twice daily for 14 days	• ↑COX • ↓C3 complement deposition in outer retina • ↓vimentin and GFAP • ↑ATP
CFH knockout mice (Calaza <i>et al.</i>) ¹⁷⁸	670 nm light via LED for 90 s	40 mW/cm ²	Daily for 5 days	
CFH knockout mice (Kokkinopoulos <i>et al.</i>) ¹⁷⁹	670 nm light via LED for 90 s	40 mW/cm ²	Four times over 2 days for 1 and 8 weeks	• Shift of C3 deposition from neural retina to RPE/BM • ↓C3b in RPE/BM. • ↓TLR 2 and TLR 4 expression inner nuclear layer • ↓calcitonin in all layers • ↓C1s, C2, C3, C4b, C3aR1, and C5r1 gene expression • ↓C3 deposition in ONL • ↓C3 expression in subretinal microglia/monocytes • ↓oxidative damage marker • 4-hydroxynonenal (4-HNE) • improvement in VA of two lines in 94.5–97% treated patients up to 36 months • ↓pigment accumulations and cystic drusen as well as • ↓metamorphopsia, ↓dyschromatopsia and ↓relative scotomas • ↓oedema and bleeding in wet AMD
Light-induced rat model of atrophic AMD (Rutar <i>et al.</i>) ¹⁸⁰	670 nm light via LED for 3 min	9 J/cm ² (50 mW/cm ²)	Daily for 5 days	• ↓C1s, C2, C3, C4b, C3aR1, and C5r1 gene expression • ↓C3 deposition in ONL • ↓C3 expression in subretinal microglia/monocytes • ↓oxidative damage marker • 4-hydroxynonenal (4-HNE) • improvement in VA of two lines in 94.5–97% treated patients up to 36 months • ↓pigment accumulations and cystic drusen as well as • ↓metamorphopsia, ↓dyschromatopsia and ↓relative scotomas • ↓oedema and bleeding in wet AMD
Prospective study of dry and wet AMD patients (Ivandic and Ivandic) ¹⁶²	780 nm light via semiconductor laser diode to the macula for 40 s	0.3 J/cm ² (7.5 mW/cm ²)	Twice weekly for 2 weeks	• improvement in VA of two lines in 94.5–97% treated patients up to 36 months • ↓pigment accumulations and cystic drusen as well as • ↓metamorphopsia, ↓dyschromatopsia and ↓relative scotomas • ↓oedema and bleeding in wet AMD
TORPA II study – interventional longitudinal case series (Merry <i>et al.</i>) ¹⁸¹	• 670 nm light via LED for 88 s • 590 and 790 nm light via LED for 35 s	• 50–80 mW/cm² (670 nm) • 4 mW/cm² (590 nm) • 0.6 mW/cm² (790 nm)	Nine sessions over 3 weeks	• ↑BCVA of 5.90 letters at 3-week treatment and 5.14 letters at 3 months • ↑contrast sensitivity at 3 weeks and 3 months at three cycles per degree • ↓Drusen volume decreased by 0.024 mm³ • ↓central drusen thickness mean of 3.78 μm

for 6 min twice daily for 2 weeks had significantly increased COX, decreased C3 complement deposition in the outer retina as well as reduced inflammatory markers vimentin and glial fibrillary acidic protein (GFAP). There were also notable changes in the morphology of RPE macrophages and dendritic cells however there was no significant change in numbers. The relatively indirect delivery of

photobiomodulation through supplemental environmental lighting in this study demonstrated the effectiveness of NIR light in penetrating tissue to reach the target. Calaza *et al.*¹⁷⁸ reported improved ATP levels in CFH mice following NIR light exposure at 40 mW/cm² supporting evidence that photobiomodulation improves oxidative phosphorylation. In a separate experiment, Kokkinopoulos¹⁷⁹ reported

that photobiomodulation can drastically ameliorate inflammation from innate immunity. CFH mice exhibited decreased C3b expression in RPE/BM following regular 90 s NIR light exposure at 40 mW/cm² over 8 weeks. Interestingly, TLR 2 and TLR 4 expression decreased in the inner nuclear but not in the in RPE layer. Expression of the systemic inflammatory marker calcitonin was also significantly in all layers. In a light-induced model of atrophic AMD, 3 min of daily NIR light exposure at 9 J/cm² (50 mW/cm²) over 5 days was reported to reduce complement propagation, C3 deposition, photoreceptor death and oxidative stress.¹⁸⁰ There have been several clinical studies investigating the effects of photobiomodulation on AMD patients including a prospective study by Ivandic and Ivandic¹⁶² where 348 eyes of 203 patients with dry and wet type AMD were exposed to 40 s of transconjunctival 780 nm light from a semiconductor laser diode with 0.3 J/cm² delivered to the macula four times over 2 weeks. The treatment group included 146 with cataracts with 182 without. The remaining 20 eyes of 10 patients received sham treatments and served as the control group. There was substantial improvement in both treated groups with 97% of patients with cataracts improving in visual acuity by a mean of two lines and 94.5% of the non-cataract patients by the same amount up to 36 months. The authors also reported reduced pigmentation and cystic drusen as well as improvement in metamorphopsia, dyschromatopsia and relative scotomas. Patients with wet AMD had reduced oedema and bleeding. No change was observed in the control group. The TORPA II study, which examined 42 eyes in 24 patients with dry AMD by Merry *et al.*¹⁸¹ consisted of delivering 88 s of 670 nm light at 4–7.7 J/cm² through LED-based devices as well as 590 and 790 nm for 35 s delivering 0.1 J/cm² in nine treatments over 3 weeks. Following photobiomodulation, there were positive functional changes indicated by improvement of contrast sensitivity and visual acuity at 3 weeks and 3 months as well as anatomical improvement exhibited by decrease in drusen volume and central drusen thickness. However, there are significant flaws in the methodologies of both studies: lack of control arm or significant number discrepancies between treatment groups, large ranges in total dosages delivered in addition to the patient demographic with varying stages of disease. Hence, it is difficult to conclude clinical efficacy of photobiomodulation in these studies. It is noteworthy to mention that in both studies there were no reported adverse effects or subjective discomfort by patients, illuminating an excellent safety profile and suggesting that NIR light treatment is well tolerated. There is also significant variance of power settings and delivery methods of photobiomodulation

between *in vitro*, animal studies and clinical studies. Thus, the optimal treatment parameters of photobiomodulation are poorly defined which poses a problem in its translation to the clinical setting.

CONCLUSION AND FUTURE DIRECTIONS

The RPE is a vital component of the eye and its dysfunction has been implicated in the pathogenesis of AMD. Photobiomodulation is a promising non-invasive therapy that has demonstrated the capacity to ameliorate oxidative stress, mitochondrial dysfunction and complement dysregulation, all of which are key mechanisms of RPE dysfunction leading to AMD.

Although clinical studies of photobiomodulation in AMD patients have reported positive functional and anatomical improvements, these studies suffer from significant flaws in methodology including the lack of a comparative control group. As a result, conclusions regarding clinical efficacy cannot be established. Hence, there is a need for randomized control trials. The LIGHTSITE study which is currently enrolling in Canada, will be a 30-subject, randomized, double-masked, sham-controlled clinical trial examining similar outcomes from the TORPA II study.¹⁸¹ As previous studies have used varying dosages, duration and frequency of NIR light with varying effectiveness, there is currently no consensus on treatment regimens. Therefore further studies are needed to establish effective treatment parameters.

REFERENCES

1. Wong WL, Su X, Li X *et al.* Global prevalence of age-related macular degeneration and disease burden projection for 2020 and 2040: a systematic review and meta-analysis. *Lancet Glob Health* 2014; **2**: e106–e16.
2. Karu TI. Cellular mechanisms of photobiomodulation. *Lasers in Dentistry: Guide for Clinical Practice*. 2015; **17**: 23–6. New Jersey, NY: Wiley and Son's.
3. Karu T. Mitochondrial mechanisms of photobiomodulation in context of new data about multiple roles of ATP. *Photomed Laser Surg* 2010; **28**: 159–60.
4. Hogan MJ, Alvarado JA, Weddell J. Ciliary body and posterior chamber. In: *Histology of the Human Eye: An Atlas and Textbook*. Philadelphia: Saunders, 1971.
5. Marshall J. The ageing retina: physiology or pathology. *Eye* 1987; **1**: 282–95.
6. Streeten BW. Development of the human retinal pigment epithelium and the posterior segment. *Arch Ophthalmol* 1969; **81**: 383–94.
7. Boulton M, Dayhaw-Barker P. The role of the retinal pigment epithelium: topographical variation and ageing changes. *Eye* 2001; **15**: 384–9.

8. Baehr W, Wu SM, Bird AC, Palczewski K. The retinoid cycle and retina disease. In: *The Retinoid Cycle and Retina Disease*, Vol. 43. Amsterdam, Netherlands: Pergamon, 2003; 2957–8.
9. Ach T, Huisinck C, McGwin G Jr *et al.* Quantitative autofluorescence and cell density maps of the human retinal pigment epithelium. *Invest Ophthalmol Vis Sci* 2014; **55**: 4832–41.
10. Ishibashi K, Tian J, Handa JT. Similarity of mRNA phenotypes of morphologically normal macular and peripheral retinal pigment epithelial cells in older human eyes. *Invest Ophthalmol Vis Sci* 2004; **45**: 3291–301.
11. Gao H, Hollyfield J. Aging of the human retina. Differential loss of neurons and retinal pigment epithelial cells. *Invest Ophthalmol Vis Sci* 1992; **33**: 1–17.
12. Del Priore LV, Kuo Y-H, Tezel TH. Age-related changes in human RPE cell density and apoptosis proportion in situ. *Invest Ophthalmol Vis Sci* 2002; **43**: 3312–8.
13. Feeney-Burns L, Hilderbrand ES, Eldridge S. Aging human RPE: morphometric analysis of macular, equatorial, and peripheral cells. *Invest Ophthalmol Vis Sci* 1984; **25**: 195–200.
14. Cunha-Vaz JG. The blood-retinal barriers. *Doc Ophthalmol* 1976; **41**: 287–327.
15. Cunha-Vaz JG. Blood—retinal barrier and new perspectives of management of retinal disease. In: Coscas GPiccolino FC, eds. *Blood—Retinal Barrier and New Perspectives of Management of Retinal Disease*. Dordrecht: Springer, 1998; 5–11.
16. Miller SS, Steinberg RH. Passive ionic properties of frog retinal pigment epithelium. *J Membr Biol* 1977; **36**: 337–72.
17. Miller SS, Steinberg RH. Active transport of ions across frog retinal pigment epithelium. *Exp Eye Res* 1977; **25**: 235–48.
18. Rizzolo LJ, Li ZQ. Diffusible, retinal factors stimulate the barrier properties of junctional complexes in the retinal pigment epithelium. *J Cell Sci* 1993; **106**: 859–67.
19. Ved N, Hulse RP, Bestall SM, Donaldson LF, Bainbridge JW, Bates DO. Vascular endothelial growth factor-a(165)b ameliorates outer-retinal barrier and vascular dysfunction in the diabetic retina. *Clin Sci (Lond)* 2017; **131**: 1225–43.
20. Desjardins DM, Yates PW, Dahrouj M, Liu Y, Crosson CE, Ablonczy Z. Progressive early breakdown of retinal pigment epithelium function in hyperglycemic rats. *Invest Ophthalmol Vis Sci* 2016; **57**: 2706–13.
21. Xiao Q, Zeng S, Ling S, Lv M. Up-regulation of HIF-1 α and VEGF expression by elevated glucose concentration and hypoxia in cultured human retinal pigment epithelial cells. *J Huazhong Univ Sci Technol Med Sci* 2006; **26**: 463–5.
22. Zhang P, Wang Y, Hui Y *et al.* Inhibition of VEGF expression by targeting HIF-1 α with small interference RNA in human RPE cells. *Ophthalmologica* 2007; **221**: 411–7.
23. Ban Y, Wilt SD, Rizzolo LJ. Two secreted retinal factors regulate different stages of development of the outer blood—retinal barrier. *Dev Brain Res* 2000; **119**: 259–67.
24. Rózanowska M, Jarvis-Evans J, Korytowski W, Boulton ME, Burke JM, Sarna T. Blue light-induced reactivity of retinal age pigment in vitro generation of oxygen-reactive species. *J Biol Chem* 1995; **270**: 18825–30.
25. Rózanowska M, Wessels J, Boulton M *et al.* Blue light-induced singlet oxygen generation by retinal lipofuscin in non-polar media. *Free Radic Biol Med* 1998; **24**: 1107–12.
26. Boulton M. Melanin and the retinal pigment epithelium. *Retinal Pigm Epithelium* 1998: 68–85.
27. DN H, Simon JD, Sarna T. Role of ocular melanin in ophthalmic physiology and pathology. *Photochem Photobiol* 2008; **84**: 639–44.
28. Frank RN, Amin RH, Puklin JE. Antioxidant enzymes in the macular retinal pigment epithelium of eyes with neovascular age-related macular degeneration. *Am J Ophthalmol* 1999; **127**: 694–709.
29. Miceli MV, Liles MR, Newsome DA. Evaluation of oxidative processes in human pigment epithelial cells associated with retinal outer segment phagocytosis. *Exp Cell Res* 1994; **214**: 242–9.
30. Newsome D, Dobard E, Liles M, Oliver P. Human retinal pigment epithelium contains two distinct species of superoxide dismutase. *Invest Ophthalmol Vis Sci* 1990; **31**: 2508–13.
31. Oliver PD, Newsome DA. Mitochondrial superoxide dismutase in mature and developing human retinal pigment epithelium. *Invest Ophthalmol Vis Sci* 1992; **33**: 1909–18.
32. Tate D, Miceli MV, Newsome DA. Phagocytosis and H₂O₂ induce catalase and metallothionein gene expression in human retinal pigment epithelial cells. *Invest Ophthalmol Vis Sci* 1995; **36**: 1271–9.
33. Kurtz JC, Jones DP, Sternberg P, Wu MW, Olsen W. Antioxidant functions of glutathione in human retinal pigment epithelium in relation to age-related macular degeneration. In: Coscas GPiccolino FC, eds. *Antioxidant Functions of Glutathione in Human Retinal Pigment Epithelium in Relation to Age-Related Macular Degeneration*. Dordrecht: Springer, 1998; 47–57.
34. Sternberg P, Davidson PC, Jones DP, Hagen TM, Reed RL, Drews-Botsch C. Protection of retinal pigment epithelium from oxidative injury by glutathione and precursors. *Invest Ophthalmol Vis Sci* 1993; **34**: 3661–8.
35. Kumagai AK, Glasgow BJ, Pardridge WM. GLUT1 glucose transporter expression in the diabetic and nondiabetic human eye. *Invest Ophthalmol Vis Sci* 1994; **35**: 2887–94.
36. Hosoya K-I, Tomi M, Ohtsuki S *et al.* Conditionally immortalized retinal capillary endothelial cell lines (TR-iBRB) expressing differentiated endothelial cell functions derived from a transgenic rat. *Exp Eye Res* 2001; **72**: 163–72.
37. Shadrach K, Senanayake P, Nishiyama K *et al.* Glucose utilization by human RPE cultures. *Invest Ophthalmol Vis Sci* 2003; **44**: 394.
38. Verlangieri AJ, Stevens JW. L-ascorbic acid: effects on aortic glycosaminoglycan 35S incorporation in rabbit-induced atherogenesis. *J Vasc Res* 1979; **16**: 177–85.

39. Kaufman SD. Coenzymes and hydroxylases: ASCORBATE and dopamine- β -hydroxylase; TETRA-HYDROPTERIDINES and phenylalanine and tyrosine hydroxylases. *Pharmacol Rev* 1966; **18**: 61–9.
40. Khatami M, Stramm LE, Rookey JH. Ascorbate transport in cultured cat retinal pigment epithelial cells. *Exp Eye Res* 1986; **43**: 607–15.
41. Bazan NG, Gordon WC, de Turco EBR. Docosahexaenoic acid uptake and metabolism in photoreceptors: retinal conservation by an efficient retinal pigment epithelial cell-mediated recycling process. In: *Neurobiology of Essential Fatty Acids*. Boston, MA: Springer, 1992; 295–306.
42. Bazan NG, de Turco EBR, Gordon WC. Docosahexaenoic acid supply to the retina and its conservation in photoreceptor cells by active retinal pigment epithelium-mediated recycling. In: *Fatty Acids And Lipids: Biological Aspects*, Vol. 75. Basel: Karger Publishers, 1994; 120–3.
43. Bazan NG, Turco EBR, Gordon WC. Pathways for the uptake and conservation of docosahexaenoic acid in photoreceptors and synapses: biochemical and autoradiographic studies. *Can J Physiol Pharmacol* 1993; **71**: 690–8.
44. Chen Y, Houghton LA, Brenna JT, Noy N. Docosahexaenoic acid modulates the interactions of the interphotoreceptor retinoid-binding protein with 11-cis-retinal. *J Biol Chem* 1996; **271**: 20507–15.
45. Hurley JB, Lindsay KJ, Du J. Glucose, lactate, and shuttling of metabolites in vertebrate retinas. *J Neurosci Res* 2015; **93**: 1079–92.
46. Adler AJ, Southwick RE. Distribution of glucose and lactate in the interphotoreceptor matrix. *Ophthalmic Res* 1992; **24**: 243–52.
47. Philp NJ, Yoon H, Grollman EF. Monocarboxylate transporter MCT1 is located in the apical membrane and MCT3 in the basal membrane of rat RPE. *Am J Phys Regul Integr Comp Phys* 1998; **274**: R1824–R8.
48. Kenyon E, Yu K, La Cour M, Miller SS. Lactate transport mechanisms at apical and basolateral membranes of bovine retinal pigment epithelium. *Am J Phys Cell Phys* 1994; **267**: C1561–C73.
49. Hamann S, Zeuthen T, La Cour M et al. Aquaporins in complex tissues: distribution of aquaporins 1–5 in human and rat eye. *Am J Phys Cell Phys* 1998; **274**: C1332–C45.
50. Stamer WD, Bok D, Hu J, Jaffe GJ, McKay BS. Aquaporin-1 channels in human retinal pigment epithelium: role in transepithelial water movement. *Invest Ophthalmol Vis Sci* 2003; **44**: 2803–8.
51. Rizzolo LJ. Polarization of the Na⁺, K⁺-ATPase in epithelia derived from the neuroepithelium. *Int Rev Cytol* 1998; **185**: 195–235.
52. Ostwald TJ, Steinberg RH. Localization of frog retinal pigment epithelium Na⁺-K⁺ ATPase. *Exp Eye Res* 1980; **31**: 351–60.
53. Rizzolo LJ. The distribution of Na⁺, K⁺-ATPase in the retinal pigmented epithelium from chicken embryo is polarized in vivo but not in primary cell culture. *Exp Eye Res* 1990; **51**: 435–46.
54. Wimmers S, Karl MO, Strauss O. Ion channels in the RPE. *Prog Retin Eye Res* 2007; **26**: 263–301.
55. Strauss O. The retinal pigment epithelium in visual function. *Physiol Rev* 2005; **85**: 845–81.
56. Weng T, Godley BF, Jin G et al. Oxidant and antioxidant modulation of chloride channels expressed in human retinal pigment epithelium. *Am J Phys Cell Phys* 2002; **283**: C839–C49.
57. Wills NK, Weng T, Mo L et al. Chloride channel expression in cultured human fetal RPE cells: response to oxidative stress. *Invest Ophthalmol Vis Sci* 2000; **41**: 4247–55.
58. Beatty S, Koh H-H, Phil M, Henson D, Boulton M. The role of oxidative stress in the pathogenesis of age-related macular degeneration. *Surv Ophthalmol* 2000; **45**: 115–34.
59. Dibas A, Yorio T. Regulation of transport in the RPE. In: *Ocular Transporters in Ophthalmic Diseases and Drug Delivery*. New York: Humana Press, 2008; 157–84.
60. Edelman JL, Lin H, Miller SS. Acidification stimulates chloride and fluid absorption across frog retinal pigment epithelium. *Am J Phys Cell Phys* 1994; **266**: C946–C56.
61. Ryeom SW, Silverstein RL, Scotto A, Sparrow JR. Binding of anionic phospholipids to retinal pigment epithelium may be mediated by the scavenger receptor CD36. *J Biol Chem* 1996; **271**: 20536–9.
62. Ryeom SW, Sparrow JR, Silverstein RL. CD36 participates in the phagocytosis of rod outer segments by retinal pigment epithelium. *J Cell Sci* 1996; **109**: 387–95.
63. Tarnowski B, Shepherd VL, McLaughlin BJ. Mannose 6-phosphate receptors on the plasma membrane on rat retinal pigment epithelial cells. *Invest Ophthalmol Vis Sci* 1988; **29**: 291–7.
64. Lin H, Clegg DO. Integrin α v β 5 participates in the binding of photoreceptor rod outer segments during phagocytosis by cultured human retinal pigment epithelium. *Invest Ophthalmol Vis Sci* 1998; **39**: 1703–12.
65. D'Cruz PM, Yasumura D, Weir J et al. Mutation of the receptor tyrosine kinase gene *Mertk* in the retinal dystrophic RCS rat. *Hum Mol Genet* 2000; **9**: 645–51.
66. Feng W, Yasumura D, Matthes MT, LaVail MM, Vollrath D. *Mertk* triggers uptake of photoreceptor outer segments during phagocytosis by cultured retinal pigment epithelial cells. *J Biol Chem* 2002; **277**: 17016–22.
67. Chang Y, Finnemann SC. Tetraspanin CD81 is required for the α v β 5-integrin-dependent particle-binding step of RPE phagocytosis. *J Cell Sci* 2007; **120**: 3053–63.
68. Karl MO, Kroeger W, Wimmers S et al. Endogenous Gas6 and Ca²⁺-channel activation modulate phagocytosis by retinal pigment epithelium. *Cell Signal* 2008; **20**: 1159–68.
69. Slomiany MG, Rosenzweig SA. Autocrine effects of IGF-I-induced VEGF and IGFBP-3 secretion in retinal pigment epithelial cell line ARPE-19. *Am J Phys Cell Phys* 2004; **287**: C746–C53.
70. Walsh N, Valter K, Stone J. Cellular and subcellular patterns of expression of bFGF and CNTF in the normal and light stressed adult rat retina. *Exp Eye Res* 2001; **72**: 495–501.
71. Campochiaro PA, Hackett SF, Viores SA et al. Platelet-derived growth factor is an autocrine growth

- stimulator in retinal pigmented epithelial cells. *J Cell Sci* 1994; **107**: 2459–69.
72. Ahuja P, Caffé A, Holmqvist I *et al.* Lens epithelium-derived growth factor (LEDGF) delays photoreceptor degeneration in explants of rd/rd mouse retina. *Neuroreport* 2001; **12**: 2951–5.
73. Adamis A, Shima D, Yeo K-T *et al.* Synthesis and secretion of vascular permeability factor/vascular endothelial growth factor by human retinal pigment epithelial cells. *Biochem Biophys Res Commun* 1993; **193**: 631–8.
74. Tombran-Tink J, Chader GG, Johnson LV. PEDF: a pigment epithelium-derived factor with potent neuronal differentiative activity. *Exp Eye Res* 1991; **53**: 411–4.
75. Tanihara H, Yoshida M, Matsumoto M, Yoshimura N. Identification of transforming growth factor-beta expressed in cultured human retinal pigment epithelial cells. *Invest Ophthalmol Vis Sci* 1993; **34**: 413–9.
76. Amin R, Puklin JE, Frank RN. Growth factor localization in choroidal neovascular membranes of age-related macular degeneration. *Invest Ophthalmol Vis Sci* 1994; **35**: 3178–88.
77. Witmer A, Vrensen G, Van Noorden C, Schlingemann R. Vascular endothelial growth factors and angiogenesis in eye disease. *Prog Retin Eye Res* 2003; **22**: 1–29.
78. Rosenthal R, Heimann H, Agostini H, Martin G, Hansen LL, Strauss O. Ca²⁺ channels in retinal pigment epithelial cells regulate vascular endothelial growth factor secretion rates in health and disease. *Mol Vis* 2007; **13**: 443.
79. Barrett DJ, Redmond TM, Wiggert B, Oprian DD, Chader GJ, Nickerson JM. cDNA clones encoding bovine interphotoreceptor retinoid binding protein. *Biochem Biophys Res Commun* 1985; **131**: 1086–93.
80. Fong S-L, Liou GI, Landers R, Alvarez R, Bridges C. Purification and characterization of a retinol-binding glycoprotein synthesized and secreted by bovine neural retina. *J Biol Chem* 1984; **259**: 6534–42.
81. Gudas L, Sporn M, Roberts A, Sporn M, Roberts A, Goodman D. In: Sporn MB, Roberts AB, Goodman DS, eds. *The Retinoids: Biology, Chemistry and Medicine*. New York: Raven Press, 1994.
82. Sun H, Kawaguchi R. The membrane receptor for plasma retinol-binding protein, a new type of cell-surface receptor. *Int Rev Cell Mol Biol* 2011; **288**: 1–41.
83. Detrick B, Hooks JJ. Immune regulation in the retina. *Immunol Res* 2010; **47**: 153–61.
84. Ferguson TA, Griffith TS. A vision of cell death: insights into immune privilege. *Immunol Rev* 1997; **156**: 167–84.
85. Perez VL, Caspi RR. Immune mechanisms in inflammatory and degenerative eye disease. *Trends Immunol* 2015; **36**: 354–63.
86. Zavazava N, Halene M, Westphal E *et al.* Expression of MHC class I and II molecules by cadaver retinal pigment epithelium cells: optimization of post-mortem HLA typing. *Clin Exp Immunol* 1991; **84**: 163–6.
87. Osusky R, Dorio RJ, Arora YK, Ryan SJ, Walker SM. MHC class II positive retinal pigment epithelial (RPE) cells can function as antigen-presenting cells for microbial superantigen. *Ocul Immunol Inflamm* 1997; **5**: 43–50.
88. Anderson DH, Radeke MJ, Gallo NB *et al.* The pivotal role of the complement system in aging and age-related macular degeneration: hypothesis re-visited. *Prog Retin Eye Res* 2010; **29**: 95–112.
89. Joachim N, Mitchell P, Rochtchina E, Tan AG, Wang JJ. Incidence and progression of reticular drusen in age-related macular degeneration: findings from an older Australian cohort. *Ophthalmology* 2014; **121**: 917–25.
90. Arnold JJ, Sarks SH, Killingsworth MC, Sarks JP. Reticular pseudodrusen. A risk factor in age-related maculopathy. *Retina* 1995; **15**: 183–91.
91. Ambati J, Fowler BJ. Mechanisms of age-related macular degeneration. *Neuron* 2012; **75**: 26–39.
92. Qin S, Rodrigues GA. Progress and perspectives on the role of RPE cell inflammatory responses in the development of age-related macular degeneration. *J Inflamm Res* 2008; **1**: 49.
93. Maeda H, Ogata N, Yi X, Takeuchi M, Ohkuma H, Uyama M. Apoptosis of photoreceptor cells in ornithine-induced retinopathy. *Graefes Arch Clin Exp Ophthalmol* 1998; **236**: 207–12.
94. Curcio CA. Photoreceptor topography in ageing and age-related maculopathy. *Eye* 2001; **15**: 376–83.
95. Spaide RF, Curcio CA, Ooto S, Suzuki M. Histological evidence of outer retinal atrophy associated with geographic atrophy secondary to age-related macular degeneration. *Invest Ophthalmol Vis Sci* 2014; **55**: 3555.
96. Aryan N, Betts-Obregon BS, Perry G, Tsin AT. Oxidative stress induces senescence in cultured RPE cells. *Open Neurol J* 2016; **10**: 83–7.
97. Chen X-D, M-Y S, Chen T-T, Hong H-Y, Han A-D, Li W-S. Oxidative stress affects retinal pigment epithelial cell survival through epidermal growth factor receptor/AKT signaling pathway. *Int J Ophthalmol* 2017; **10**: 507.
98. Jin G-F, Hurst JS, Godley BF. Hydrogen peroxide stimulates apoptosis in cultured human retinal pigment epithelial cells. *Curr Eye Res* 2001; **22**: 165–73.
99. Hinton DR, He S, Lopez PF. Apoptosis in surgically excised choroidal neovascular membranes in age-related macular degeneration. *Arch Ophthalmol* 1998; **116**: 203–9.
100. Qin S. Oxidative damage of retinal pigment epithelial cells and age-related macular degeneration. *Drug Dev Res* 2007; **68**: 213–25.
101. Shen J, Dong A, Hackett S, Bell W, Green W, Campochiaro PA. Oxidative damage in age-related macular degeneration. *Histol Histopathol* 2007; **22**: 1301–8.
102. Imamura Y, Noda S, Hashizume K *et al.* Drusen, choroidal neovascularization, and retinal pigment epithelium dysfunction in SOD1-deficient mice: a model of age-related macular degeneration. *Proc Natl Acad Sci* 2006; **103**: 11282–7.
103. Gu X, Meer SG, Miyagi M *et al.* Carboxyethylpyrrole protein adducts and autoantibodies, biomarkers for age-related macular degeneration. *J Biol Chem* 2003; **278**: 42027–35.

104. Hollyfield JG, Bonilha VL, Rayborn ME *et al.* Oxidative damage-induced inflammation initiates age-related macular degeneration. *Nat Med* 2008; **14**: 194–8.
105. Ebrahim Q, Renganathan K, Sears J *et al.* Carboxyethylpyrrole oxidative protein modifications stimulate neovascularization: implications for age-related macular degeneration. *Proc Natl Acad Sci* 2006; **103**: 13480–4.
106. Brunk UT, Terman A. Lipofuscin: mechanisms of age-related accumulation and influence on cell function. *Free Radic Biol Med* 2002; **33**: 611–9.
107. Sparrow JR, Nakanishi K, Parish CA. The lipofuscin fluorophore A2E mediates blue light-induced damage to retinal pigmented epithelial cells. *Invest Ophthalmol Vis Sci* 2000; **41**: 1981–9.
108. Sparrow JR, Cai B. Blue light-induced apoptosis of A2E-containing RPE: involvement of caspase-3 and protection by Bcl-2. *Invest Ophthalmol Vis Sci* 2001; **42**: 1356–62.
109. Ablonczy Z, Higbee D, Anderson DM *et al.* Lack of correlation between the spatial distribution of A2E and lipofuscin fluorescence in the human retinal pigment epithelium. *Invest Ophthalmol Vis Sci* 2013; **54**: 5535–42.
110. Adler L, Boyer NP, Anderson DM *et al.* Determination of N-retinylidene-N-retinylethanolamine (A2E) levels in central and peripheral areas of human retinal pigment epithelium. *Photochem Photobiol Sci* 2015; **14**: 1983–90.
111. Gliem M, Müller PL, Finger RP, McGuinness MB, Holz FG, Charbel Issa P. Quantitative fundus autofluorescence in early and intermediate age-related macular degeneration. *JAMA Ophthalmol* 2016; **134**: 817–24.
112. Chan D. Cigarette smoking and age-related macular degeneration. *Optom Vis Sci* 1998; **75**: 476–84.
113. Wang AL, Lukas TJ, Yuan M, Du N, Handa JT, Neufeld AH. Changes in retinal pigment epithelium related to cigarette smoke: possible relevance to smoking as a risk factor for age-related macular degeneration. *PLoS One* 2009; **4**: e5304.
114. Group A-REDSR. A randomized, placebo-controlled, clinical trial of high-dose supplementation with vitamins C and E, beta carotene, and zinc for age-related macular degeneration and vision loss: AREDS report no. 8. *Arch Ophthalmol* 2001; **119**: 1417.
115. Chew EY, Clemons TE, SanGiovanni JP *et al.* Lutein + zeaxanthin and omega-3 fatty acids for age-related macular degeneration: the age-related eye disease study 2 (AREDS2) randomized clinical trial. *JAMA* 2013; **309**: 2005–15.
116. King A, Gottlieb E, Brooks DG, Murphy MP, Dunaief JL. Mitochondria-derived reactive oxygen species mediate blue light-induced death of retinal pigment epithelial cells. *Photochem Photobiol* 2004; **79**: 470–5.
117. Ballinger SW, Van Houten B, Conklin CA, Jin G-F, Godley BF. Hydrogen peroxide causes significant mitochondrial DNA damage in human RPE cells. *Exp Eye Res* 1999; **68**: 765–72.
118. Lin H, Xu H, Liang F-Q *et al.* Mitochondrial DNA damage and repair in RPE associated with aging and age-related macular degeneration. *Invest Ophthalmol Vis Sci* 2011; **52**: 3521–9.
119. Nordgaard CL, Karunadharma PP, Feng X, Olsen TW, Ferrington DA. Mitochondrial proteomics of the retinal pigment epithelium at progressive stages of age-related macular degeneration. *Invest Ophthalmol Vis Sci* 2008; **49**: 2848–55.
120. Lamb LE, Simon JD. A2E: a component of ocular lipofuscin. *Photochem Photobiol* 2004; **79**: 127–36.
121. Suter M, Remé C, Grimm C *et al.* Age-related macular degeneration the lipofuscin componentn-retinyl-n-retinylidene ethanolamine detaches proapoptotic proteins from mitochondria and induces apoptosis in mammalian retinal pigment epithelial cells. *J Biol Chem* 2000; **275**: 39625–30.
122. Wielgus A, Collier R, Martin E *et al.* Blue light induced A2E oxidation in rat eyes—experimental animal model of dry AMD. *Photochem Photobiol Sci* 2010; **9**: 1505–12.
123. Karunadharma PP, Nordgaard CL, Olsen TW, Ferrington DA. Mitochondrial DNA damage as a potential mechanism for age-related macular degeneration. *Invest Ophthalmol Vis Sci* 2010; **51**: 5470–9.
124. Kanda A, Chen W, Othman M *et al.* A variant of mitochondrial protein LOC387715/ARMS2, not HTRA1, is strongly associated with age-related macular degeneration. *Proc Natl Acad Sci* 2007; **104**: 16227–32.
125. Fritsche LG, Loenhardt T, Janssen A *et al.* Age-related macular degeneration is associated with an unstable ARMS2 (LOC387715) mRNA. *Nat Genet* 2008; **40**: 892–6.
126. Michaelides M, Jenkins SA, Bamiou D *et al.* Macular dystrophy associated with the a3243g mitochondrial dna mutation: distinct retinal and associated features, disease variability, and characterization of asymptomatic family members. *Arch Ophthalmol* 2008; **126**: 320–8.
127. Daruich A, Matet A, Borruat F-X. Macular dystrophy associated with the mitochondrial DNA A3243G mutation: pericentral pigment deposits or atrophy? Report of two cases and review of the literature. *BMC Ophthalmol* 2014; **14**: 77.
128. Gold B, Merriam JE, Zernant J *et al.* Variation in factor B (BF) and complement component 2 (C2) genes is associated with age-related macular degeneration. *Nat Genet* 2006; **38**: 458–62.
129. Ennis S, Jomary C, Mullins R *et al.* Association between the SERPING1 gene and age-related macular degeneration: a two-stage case-control study. *Lancet* 2008; **372**: 1828–34.
130. Park KH, Ryu E, Tosakulwong N, Wu Y, Edwards AO. Common variation in the SERPING1 gene is not associated with age-related macular degeneration in two independent groups of subjects. *Mol Vis* 2009; **15**: 200.
131. Klein RJ, Zeiss C, Chew EY *et al.* Complement factor H polymorphism in age-related macular degeneration. *Science* 2005; **308**: 385–9.
132. Haines JL, Hauser MA, Schmidt S *et al.* Complement factor H variant increases the risk of age-related macular degeneration. *Science* 2005; **308**: 419–21.

133. Edwards AO, Ritter R, Abel KJ, Manning A, Panhuysen C, Farrer LA. Complement factor H polymorphism and age-related macular degeneration. *Science* 2005; **308**: 421–4.
134. Ennis S, Gibson J, Cree AJ, Collins A, Lotery AJ. Support for the involvement of complement factor I in age-related macular degeneration. *Eur J Hum Genet* 2010; **18**: 15–16.
135. Fagerness JA, Maller JB, Neale BM, Reynolds RC, Daly MJ, Seddon JM. Variation near complement factor I is associated with risk of advanced AMD. *Eur J Hum Genet* 2009; **17**: 100–4.
136. Yates JR, Sepp T, Matharu BK *et al.* Complement C3 variant and the risk of age-related macular degeneration. *N Engl J Med* 2007; **357**: 553–61.
137. Hughes AE, Orr N, Esfandiary H, Diaz-Torres M, Goodship T, Chakravarthy U. A common CFH haplotype, with deletion of CFHR1 and CFHR3, is associated with lower risk of age-related macular degeneration. *Nat Genet* 2006; **38**: 1173–7.
138. Weber BHF, Vogt G, Pruett RC, Stohr H, Felbor U. Mutations in the tissue inhibitor of metalloproteinases-3 (TIMP3) in patients with Sorsby's fundus dystrophy. *Nat Genet* 1994; **8**: 352–6.
139. Fernandez-Godino R, Garland DL, Pierce EA. A local complement response by RPE causes early-stage macular degeneration. *Hum Mol Genet* 2015; **24**: 5555–69.
140. Coffey PJ, Gias C, McDermott CJ *et al.* Complement factor H deficiency in aged mice causes retinal abnormalities and visual dysfunction. *Proc Natl Acad Sci* 2007; **104**: 16651–6.
141. Ding J-D, Kelly U, Landowski M *et al.* Expression of human complement factor H prevents age-related macular degeneration-like retina damage and kidney abnormalities in aged Cfh knockout mice. *Am J Pathol* 2015; **185**: 29–42.
142. Yang P, Baciou P, Kerrigan BCP *et al.* Retinal pigment epithelial cell death by the alternative complement Cascade: role of membrane regulatory proteins, calcium, PKC, and oxidative Stress Complement-mediated RPE cell death. *Invest Ophthalmol Vis Sci* 2014; **55**: 3012–21.
143. Bora NS, Kaliappan S, Jha P *et al.* Complement activation via alternative pathway is critical in the development of laser-induced choroidal neovascularization: role of factor B and factor H. *J Immunol* 2006; **177**: 1872–8.
144. Nozaki M, Raisler BJ, Sakurai E *et al.* Drusen complement components C3a and C5a promote choroidal neovascularization. *Proc Natl Acad Sci USA* 2006; **103**: 2328–33.
145. Bora PS, Sohn J-H, Cruz JM *et al.* Role of complement and complement membrane attack complex in laser-induced choroidal neovascularization. *J Immunol* 2005; **174**: 491–7.
146. Rohrer B, Long Q, Coughlin B *et al.* A targeted inhibitor of the alternative complement pathway reduces angiogenesis in a mouse model of age-related macular degeneration. *Invest Ophthalmol Vis Sci* 2009; **50**: 3056–64.
147. Fukuoka Y, Strainic M, Medof ME. Differential cytokine expression of human retinal pigment epithelial cells in response to stimulation by C5a. *Clin Exp Immunol* 2003; **131**: 248–53.
148. Joseph K, Kulik L, Coughlin B *et al.* Oxidative stress sensitizes retinal pigmented epithelial (RPE) cells to complement-mediated injury in a natural antibody-, lectin pathway-, and phospholipid epitope-dependent manner. *J Biol Chem* 2013; **288**: 12753–65.
149. Thurman JM, Renner B, Kunchithapautham K *et al.* Oxidative stress renders retinal pigment epithelial cells susceptible to complement-mediated injury. *J Biol Chem* 2009; **284**: 16939–47.
150. Chen M, Forrester JV, Xu H. Synthesis of complement factor H by retinal pigment epithelial cells is down-regulated by oxidized photoreceptor outer segments. *Exp Eye Res* 2007; **84**: 635–45.
151. Lau L-I, Chiou S-H, Liu CJ-L, Yen M-Y, Wei Y-H. The effect of photo-oxidative stress and inflammatory cytokine on complement factor H expression in retinal pigment epithelial cells. *Invest Ophthalmol Vis Sci* 2011; **52**: 6832–41.
152. Geneva II. Photobiomodulation for the treatment of retinal diseases: a review. *Int J Ophthalmol* 2016; **9**: 145.
153. Mester E, Spiry T, Szende B, Tota JG. Effect of laser rays on wound healing. *Am J Surg* 1971; **122**: 532–5.
154. Chung H, Dai T, Sharma SK, Huang YY, Carroll JD, Hamblin MR. The nuts and bolts of low-level laser (light) therapy. *Ann Biomed Eng* 2012; **40**: 516–33.
155. Albarracin R, Eells J, Valter K. Photobiomodulation protects the retina from light-induced photoreceptor degeneration. *Invest Ophthalmol Vis Sci* 2011; **52**: 3582–92.
156. Qu C, Cao W, Fan Y, Lin Y. Near-infrared light protect the photoreceptor from light-induced damage in rats. In: *Retinal Degenerative Diseases*. New York: Springer, 2010; 365–74.
157. Begum R, Powner MB, Hudson N, Hogg C, Jeffery G. Treatment with 670 nm light up regulates cytochrome C oxidase expression and reduces inflammation in an age-related macular degeneration model. *PLoS One* 2013; **8**: e7828.
158. Kirdl DK, Gopalakrishnan S, Schmitt H *et al.* Photobiomodulation reduces photoreceptor death and regulates cytoprotection in early states of P23H retinal dystrophy. *Proc SPIE* 2013; **8569**: 85690F-1.
159. Tang J, Du Y, Lee CA, Talahalli R, Eells JT, Kern TS. Low-intensity far-red light inhibits early lesions that contribute to diabetic retinopathy: in vivo and in Vitro Far-red light inhibits DR. *Invest Ophthalmol Vis Sci* 2013; **54**: 3681–90.
160. Natoli R, Valter K, Barbosa M *et al.* 670nm photobiomodulation as a novel protection against retinopathy of prematurity: evidence from oxygen induced retinopathy models. *PLoS One* 2013; **8**: e72135.
161. Albarracin R, Natoli R, Rutar M, Valter K, Provis J. 670 nm light mitigates oxygen-induced degeneration in C57BL/6J mouse retina. *BMC Neurosci* 2013; **14**: 125.
162. Ivandic BT, Ivandic T. Low-level laser therapy improves vision in patients with age-related macular degeneration. *Photomed Laser Surg* 2008; **26**: 241–5.

163. Merry G, Dotson R, Devenyi R, Markowitz S, Reyes S. Photobiomodulation as a new treatment for dry age related macular degeneration. Results from the toronto and oak ridge photobimodulation study in AMD (TORPA). *Invest Ophthalmol Vis Sci* 2012; **53**: 2049.
164. Tang J, Herda AA, Kern TS. Photobiomodulation in the treatment of patients with non-center-involving diabetic macular oedema. *Br J Ophthalmol* 2014; **98**: 1013–15.
165. Ivandic BT, Ivandic T. Low-level laser therapy improves visual acuity in adolescent and adult patients with amblyopia. *Photomed Laser Surg* 2012; **30**: 167–71.
166. Ivandic BT, Ivandic T. Low-level laser therapy improves vision in a patient with retinitis pigmentosa. *Photomed Laser Surg* 2014; **32**: 181–4.
167. Eells J, Henry M, Summerfelt P et al. Therapeutic photobiomodulation for methanol-induced retinal toxicity. *Proc Natl Acad Sci USA* 2003; **100**: 3439–44.
168. Wong-Riley MT, Liang HL, Eells JT et al. Photobiomodulation directly benefits primary neurons functionally inactivated by toxins role of cytochrome c oxidase. *J Biol Chem* 2005; **280**: 4761–71.
169. Desmettre T, Maurage CA, Mordon S. Transpupillary thermotherapy (TTT) with short duration laser exposures induce heat shock protein (HSP) hyperexpression on choroidoretinal layers. *Lasers Surg Med* 2003; **33**: 102–7.
170. Karu TI, Pyatibrat LV, Kolyakov SF, Afanasyeva NI. Absorption measurements of a cell monolayer relevant to phototherapy: reduction of cytochrome c oxidase under near IR radiation. *J Photochem Photobiol B Biol* 2005; **81**: 98–106.
171. de Freitas LF, Hamblin MR. Proposed mechanisms of photobiomodulation or low-level light therapy. *IEEE J Sel Top Quantum Electron* 2016; **22**: 348–64.
172. Huang YY, Nagata K, Tedford CE, McCarthy T, Hamblin MR. Low-level laser therapy (LLLT) reduces oxidative stress in primary cortical neurons in vitro. *J Biophotonics* 2013; **6**: 829–38.
173. Huang Y-Y, Sharma SK, Carroll J, Hamblin MR. Biphasic dose response in low level light therapy – an update. *Dose-Response* 2011; **9**: 602–618. dose-response.11-009.Hamblin, dose, response.1.
174. Lavey BJ, Estlack LE, Schuster KJ, Rockwell BA, Wigle JC. The response of human retinal pigmented epithelial cells in vitro to changes in nitric oxide concentration stimulated by low levels of red light. *Proc SPIE* 2013; **8569**: 85690Q-1.
175. Fuma S, Murase H, Kuse Y, Tsuruma K, Shimazawa M, Hara H. Photobiomodulation with 670 nm light increased phagocytosis in human retinal pigment epithelial cells. *Mol Vis* 2015; **21**: 883.
176. Schuster K, Estlack L, Wigle J. Exposing human retinal pigmented epithelial cells to red light in vitro elicits an adaptive response to a subsequent 2- μ m laser challenge. In: *Mechanisms for Low-Light Therapy VIII*, Vol. **8569**. International Society for Optics and Photonics, 2013; 856908–10.
177. Kokkinopoulos I, Colman A, Hogg C, Heckenlively J, Jeffery G. Age-related retinal inflammation is reduced by 670 nm light via increased mitochondrial membrane potential. *Neurobiol Aging* 2013; **34**: 602–9.
178. Calaza KC, Kam JH, Hogg C, Jeffery G. Mitochondrial decline precedes phenotype development in the complement factor H mouse model of retinal degeneration but can be corrected by near infrared light. *Neurobiol Aging* 2015; **36**: 2869–76.
179. Kokkinopoulos I. 670nm LED ameliorates inflammation in the CFH-/- mouse neural retina. *J Photochem Photobiol B Biol* 2013; **122**: 24–31.
180. Rutar M, Natoli R, Albarracin R, Valter K, Provis J. 670-nm light treatment reduces complement propagation following retinal degeneration. *J Neuroinflammation* 2012; **9**: 257.
181. Merry GF, Munk MR, Dotson RS, Walker MG, Devenyi RG. Photobiomodulation reduces drusen volume and improves visual acuity and contrast sensitivity in dry age-related macular degeneration. *Acta Ophthalmol* 2017; **95**: e270–7.